EASTERN UNIVERSITY

ENGINEERING DRAWING LAB MANUAL

COURSE CODE: 07322110/CEN 210

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List of Instruments:

1. Drawing Board
2. T-Square
3. Set- Square
4. Drawing Templates
5. Compass for Engineering Drawing
6. Lead Pencils
7. Eraser
8. Sharpener
9. Sand Paper
10.Adhesive Tape
11.Pointer (0.1 mm)

General Instructions for all drawing sheets:

1) Bring all the mentioned equipment in each lab session.
2) Clean the drawing board by using the duster.
3) Place the drawing sheet correctly straight on the drawing board by the help of T-square.
4) Paste the drawing sheet on the drawing board by suing adhesive tape or drawing pins, make sure the adhesive tape or pins are only 10 mm away from each corner and side of drawing sheet.
5) Before any drawing make sure the drawing pencil is sharp and during drawing any line keep the pencil rotating so that the thickness of line remain same.
6) Draw the boarders on the drawing sheet 20 mm from the left side and 10 mm from top, bottom and right sides.
7) Draw the title block on the right bottom corner of sheet, having length 180 mm and height 65 mm, and divide the title block into 6 horizontal rows, each having 10 mm spacing.
8) Measure the total length and width of the drawing sheet and the object which is required to be drawn, within the boarders drawn.
9) Calculate the horizontal margins for all orthographic drawings by subtracting the total length + Width of the object from the total length of the sheet and dividing it by 3, similarly in case of vertical margins the height + Width of object will be subtracted from the height of sheet and divided by 3.
10) For erasing any unnecessary or mistakenly drawn lines use erasing shield.
11) Use duster for removing any rubber residues, left on the sheet rather using your hands.
12) Use drawing template for drawing any circle having radius smaller than 5 mm.
13) After completing the drawing carefully remove the tape or pins and store it safely in file.

LAB EXPERIMENT-1

OBJECTIVE

Introduction to Engineering Drawing.

BRIEF DISCUSSION AND EXPLANATION

What is an engineering drawing?

An engineering drawing is a subcategory of technical drawings that show the shape, structure, dimensions, tolerances, accuracy and other requirements needed to manufacture a product or part. Engineering drawings are also known as mechanical drawings, manufacturing blueprints and drawings. An engineering drawing helps to define the requirements of an engineering part and conveys the design concept. The Engineer distributes these drawings to a manufacturing department to produce the parts, to an assembly department to put the parts together, then to any vendors, archives or other company departments. The drawings can include an item or system's geometry, dimensions, tolerances, functions, finish, hardware and material.

Basic components of engineering drawing

Technical drawings may have different components depending on the industry, the part and the purpose of the drawing.

Types of lines in engineering drawings

An engineering drawing has many different kinds of lines that indicate different concepts and design ideas. Here's a list of the types of lines you can find in an engineering drawing:

Continuous/drawing line:

A solid line is the most common type of line and represents an object's physical boundaries. These are the lines used to draw objects. The line thickness can vary—thicker lines are used on the outside contour and thinner lines are used on the inner contour.

Hidden line:

Lines that are not visible in the current view are represented by dashes. They represent the edges where surfaces meet but are not directly visible. Hidden lines tend to be omitted from drawings unless they are needed to make the drawing clear.

Center line

Parts with holes and symmetrical features can be shown by using center lines. A center line indicates a circular feature on a drawing and is characterized by its long-short-long alternating line pattern. They can be used to represent:

1 Symmetry

2 Paths of motion

3 Centers of circles and the axes of symmetrical parts

Dimension and extension lines

1 Dimension and extension tines are thin lines used to indicate the sizes of features on a drawing.

2 Dimension lines show the orientation and extent of a specified length or size of an element. They end with arrowheads that terminate at the extension lines.

3 Extension lines are used to clarify the points at which a dimension begins and ends and extend toward the element that is dimensioned.

Break line

Break lines show where an object is broken to save drawing space or reveal interior features. Break lines come in two forms:

A freehand thick line

A long, ruled thin line with zigzags

Phantom line

Phantom lines are thin long-short-short-long lines used to show the travel or movement of an object or a part in alternate positions. They can also be used to show adjacent objects or features.

Section line

Section lines are used to show the cut surfaces of an object in section views. They are fine, dark lines and are generally drawn at a 45° angle. The type of line can indicate various materials such as steel, copper and brass.

Cutting plane line

Cutting plane lines are heavy lines used in section drawings to show the locations of cutting planes. When using a cutout view, the cutting plane lines show the path of the cutout.

Leader line

Leader lines are thin lines used to point to an area of a drawing requiring a note for explanation. They are usually drawn at 45-degree angles.

Types of views in engineering drawings

Engineering drawings often utilize different types of views to contribute to the understanding of a design. Here are various types of views used in engineering drawings.

Isometric view

Isometric drawings show parts as three-dimensional. Vertical lines stay vertical (compared to the front view) and parallel lines are shown at an angle, usually 30 degrees. In an isometric drawing, the object appears as if it is being viewed from above from one corner The lines that are vertical and parallel are in their true length. This means you can use a ruler and the scaling of the drawing to measure the length.

Orthographic view

An orthographic view represents a 3-D object using several two-dimensional views of the object. Standard practice calls for three orthographic views, a front, top and side view. This kind of representation allows for avoiding any kind of distortion of lengths. Different areas of the world use different angle projections to show orthographic views. You can tell which angle projection is used by the symbol shown on the drawing.

First-angle projection:

The International Organization for Standardization (ISO) calls for this kind of projection.

Third-angle projection: The American National Standards Institute (ANSI) calls for this kind of projection and is the accepted method in the United States.

Section view

A section view looks provides a view of inside an object. Sections are used to clarify the interior construction of a part that cannot be clearly described by hidden lines in exterior views. The cut material is indicated with diagonal section lines.

Cutout view:

This view is similar to the section view. While the section view shows the cutting of the entire model, and the cut-out view would be cutting only a smaller portion of the model. This view helps to reduce the number of orthographic views in the drawing.

Detail view

Detailed views are scaled-up versions of orthographic views, which are essential in complex models with small intricacies. They zoom into a selected section of a larger view.

Auxiliary view

An auxiliary view is an orthographic view for non horizontal or non vertical planes. It helps to present inclined surfaces without distortion.

Dimensions

The text of dimensions can be placed inside or outside the extension lines or drawn at an angle if space is limited. All dimensions and note text should be oriented to read from the bottom of the drawing. This is called unidirectional dimensioning.

Information blocks

These are little boxes present at the bottom corner of the engineering drawing. The block includes part name and number, author's name, coating, quantity, scale and other information.

Title block: This is usually at the bottom right of the drawing and includes identifying information like the title, number, part numbers, measurements, intellectual property notes and information about the agency that made the drawing.

Revisions block: The revisions block is usually at the top right or with the title block and lists other versions of the drawing.

Bill of materials block: This block is a list of material requirements for a given project. If a drawing requires too many items to fit in a block, then the list of materials will be provided on separate sheet.

DESIGNATION OF SHEETS:

Designation	Trimmed	Untrimmed Size
A1	594 x 841	625 x 880
A2	420 x 594	450 x 625
A3	297 x 420	330 x 450
A4	210 x 297	240 x 330

LAB EXPERIMENT-2

OBJECTIVE:

To study Orthographic Projections

BRIEF DISCUSSION AND EXPLANATION:

Orthographic Projection is a method of producing dimensioned working drawings or blueprints of 3-D Objects using a series of related 2-D views of the object to communicate the object's length, width and depth.

The views are produced by using the fundamental concept of Orthographic Projection - the location of the Spectator (the viewer). The Spectator is always located at an infinite distance from the object and planes of reference - this means that the lines of sight (projection lines) remain parallel and will project on to a Plane which is perpendicular to the projection lines.

The primary views used are called the Elevation, Plan and End Elevation and are produced by projecting an image of the object as viewed by a spectator standing at infinity on to the Planes of Reference which are then folded flat to produce a 2-D drawing. Drawings can be used using one of two methods - First Angle Projection (Used in Europe, Asia & Africa) or Third Angle Projection (Used in the USA). First Angle Projection involves the projection of the image of the object on to a Plane positioned behind the object while Third Angle projects the image on to a plane located between the object and the spectator. The method of projection alters the layout of the drawing as displayed in the image below. Drawings created using First Angle Projection will have the Plan View located below the elevation and End Elevation positioned on the side opposite the viewing direction (i.e.when viewing the left of the object the view is positioned on the right of the Elevation) while the opposite is the case for Third Angle Projection.

Conventions of Orthographic Projection

There are a number of rules and conventions which must be adhered to when producing Orthographic Drawings**:**

- Heights of objects will remain the same between Elevations including End Elevations and Auxiliary Elevations.
- Widths of Objects will remain the same between the main Elevation, Plan and Auxiliary Plans.
- Lines & Sufaces parallel to the Vertical Plane will appear as true lengths/shapes in the Elevation.
- Lines & Sufaces parallel to the Horizontal Plane will appear as true lengths/shapes in the Plan.
- 45° Lines or Arcs should be used to transfer widths between the plan and End Elevation.
- Construction lines should be drawn lightly using a H Pencil.
- Finished lines should be drawn heavily using a B Pencil.

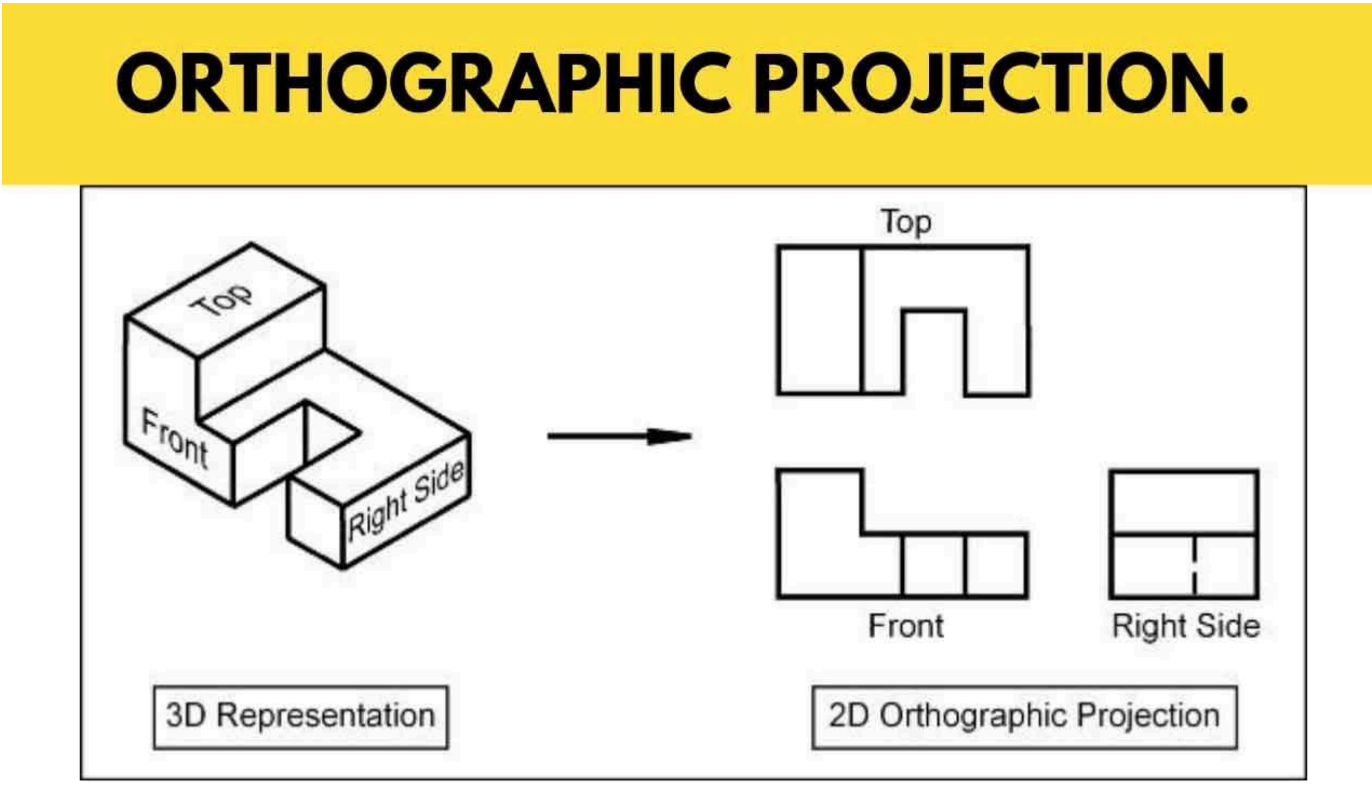


Fig: Orthographic Projection

LAB EXPERIMENT-3

OBJECTIVE

To study Isometric Projections

BRIEF DISCUSSION AND EXPLANATION

Isometric drawing, sometimes called isometric projection, is a type of 2D drawing used to draw 3D objects that is set out using 30-degree angles. It's also a type of axonometric drawing, meaning that the same scale is used for every axis, resulting in a non-distorted image. Since isometric grids are pretty easy to set up, once you understand the basics of isometric drawing, creating a freehand isometric sketch is relatively simple.

An isometric drawing is a 3D representation of an object, room, building or design on a 2D surface. One of the defining characteristics of an isometric drawing, compared to other types of 3D representation, is that the final image is not distorted and is always to scale. This is due to the fact that the foreshortening of the axes is equal.The word isometric comes from Greek to mean 'equal measure'.

Isometric drawings are a good way to show measurements and how components fit together, and is used in technical drawing, often by engineers and architects. They differ from other types of axonometric drawing, including dimetric and trimetric projections, in which different scales are used for different axes to give a distorted final image.

In an isometric drawing, the object appears as if it is being viewed from above from one corner, with the axes set out from this corner point. Isometric drawings begin with one vertical line along which two points are defined. Any lines set out from these points should be constructed at an angle of 30 degrees.

ISOMETRIC DRAWING VS ONE-POINT PERSPECTIVE

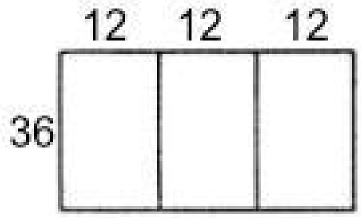
Both isometric drawings and one-point perspective drawings use geometry and mathematics to present 3D representations on 2D surfaces. One-point perspective drawings mimic the human eye, so objects appear smaller the further away they are from the viewer. In contrast, isometric drawings use parallel projection, which means objects remain at the same size, no matter how far away they are.

Basically, isometric drawing doesn’t use perspective in its rendering (i.e. lines don’t converge as they move away from the viewer). Isometric drawings are more useful for functional drawings that are used to explain how something works, while one-point perspective drawings are typically used to give a more sensory idea of an object or space.

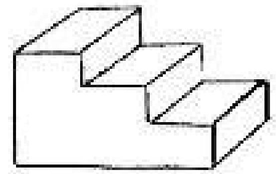
The limitation of isometric drawings compared to 3D models is that you can't change your vantage point, you have to see the drawing from the top viewpoint**.**

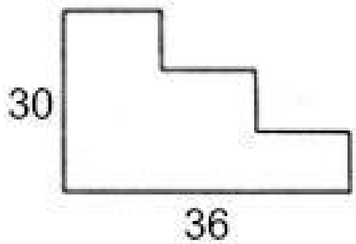
THE BENEFITS OF ISOMETRIC DRAWING

Isometric drawings are very useful for designers – particularly architects, industrial and interior designers and engineers, as they are ideal for visualising rooms, products, and infrastructure. They're a great way to quickly test out different design ideas. They also illustrate the 3D nature of an object, without being drawn in 3D software, and measurements can be made to scale along the principal axes.

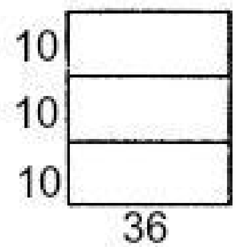


Top





Front



Right Side

LAB EXPERIMENT-4

OBJECTIVE

To study Auto CAD and various command.

BRIEF DISCUSSION AND EXPLANATION

AutoCAD is referred to as a commercial computer-aided design (CAD) and drafting software tool. It is widely utilized across an array of industries by architects, engineers, project managers and several other professionals.

AutoCAD is known term when it comes to 3D and 2D CAD design. AutoCAD software is a tool that facilitates the process of designing and communicating the end results with others. In this age most structures that are built are based on CAD drawings. The majority of them use AutoCAD.

Most people associate AutoCAD with architectural house design; however, there are many different industries that use AutoCAD. For example, many: geographical, mechanical, electronic, AEC, multimedia, electrical, surveying, engineering, town planning, and even garment companies' use AutoCAD.
Building Information Modeling (BIM) features have also been recently added to the list of things that AutoCAD can do.

AutoCAD Features*:*

AutoCAD 2D has many features and capabilities. It obviously can draw and modify structures accurately. Text and precise dimensions can be added easily. The geometry can be viewed at different scales, and the project can be organized by layers, colors, and styles. Another nice feature is that a variety of layouts can be set up for printing purposes.
Another plus is that it is easy to collaborate with other AutoCAD users over the internet. There are a variety of AutoCAD versions available on internet you can go through but most recommended is always the latest one.

Getting started with AutoCAD is relatively simple. There are three basic steps that you will need to take:

- Drawing
- Adding text and dimensions
- Printing or "plotting"

AutoCAD is made simple by the fact that it is a procedural program, which means that the command window prompts you to enter commands in a logical step by step manner.
Most architectural software has proper capability for high quality visualization and even walk through movies can be made in these software's but this is often outsourced and with a little efforts this work can be done in-house which will save the business profits and give you the chance to increase your quality

Key Features and Applications:

- **2D and 3D Drafting:**

AutoCAD allows users to create precise 2D and 3D drawings, sketches, and models.

Design and Modeling:

It's used for designing and modeling various objects, including solids, surfaces, and mesh objects.

Documentation:

AutoCAD includes features for documenting designs with tables, annotations, and more.

Automation:

It offers features to automate tasks and increase productivity, such as comparing drawings, counting objects, and adding tables.

Industry-Specific Toolsets:

AutoCAD comes with specialized toolsets for different industries, including electrical design, plant design, architectural layout, and mechanical design, according to Autodesk.

Collaboration:

Users can collaborate on designs and share them across various devices, including desktops, web, and mobile devices.

Who Uses AutoCAD?

Professionals:

Architects, engineers, designers, project managers, real estate developers, and construction professionals.

Students:

AutoCAD is also used by students learning CAD software and design principles.

Here is a list of common AutoCAD commands, categorized for easier understanding:

Drawing Commands:

- **LINE (L):** Creates straight line segments.
- **CIRCLE (C):** Creates a circle.
- **ARC (A):** Creates an arc.
- **RECTANGLE (REC):** Creates a rectangle.
- **POLYGON (POL):** Creates a regular polygon.
- **ELLIPSE (EL):** Creates an ellipse or elliptical arc.
- **PLINE (PL):** Creates a polyline (a connected sequence of line and arc segments).
- **MTEXT (MT):** Creates multiline text.
- **HATCH (H or BH):** Fills an enclosed area with a hatch pattern or solid fill.
- **GRADIENT (GD):** Fills an enclosed area with a color gradient.
- **POINT (PO):** Creates a point object.
- **SPLINE (SPL):** Creates a smooth curve (spline).
- **CONE (CONE):** Creates a 3D solid cone.
- **CYLINDER (CYL):** Creates a 3D solid cylinder.
- **BOX (BOX):** Creates a 3D solid box.
- **SPHERE (SPH):** Creates a 3D solid sphere.

- **TORUS (TOR):** Creates a 3D solid torus (donut shape).
- **WEDGE (WEDGE):** Creates a 3D solid wedge.

Editing Commands:

- **ERASE (E):** Removes objects from the drawing.
- **COPY (CO or CP):** Creates copies of selected objects.
- **MOVE (M):** Moves selected objects.
- **ROTATE (RO):** Rotates selected objects.
- **SCALE (SC):** Changes the size of selected objects.
- **STRETCH (S):** Stretches portions of selected objects.
- **TRIM (TR):** Trims objects to meet the edges of other objects.
- **EXTEND (EX):** Extends objects to meet the edges of other objects.
- **OFFSET (O):** Creates concentric circles, parallel lines, or parallel curves.
- **FILLET (F):** Creates rounded corners between two objects.
- **CHAMFER (CHA):** Creates beveled edges between two objects.
- **ARRAY (AR):** Creates multiple copies of objects in a pattern (rectangular, polar, or path).
- **MIRROR (MI):** Creates a mirrored copy of selected objects.
- **BREAK (BR):** Breaks an object into two separate objects.
- **JOIN (J):** Joins collinear lines, arcs, and polylines into a single object.
- **EXPLODE (X):** Breaks a compound object (like a block or polyline) into its individual components.
- **OFFSETEDGE (OFFSETEDGE):** Creates a parallel surface by offsetting a face or edges of a 3D solid or surface.
- **UNION (UNION):** Combines selected 3D solids or regions into a single, unified object.
- **SUBTRACT (SU):** Subtracts one set of 3D solids or regions from another.
- **INTERSECT (IN):** Creates a 3D solid or region from the intersection of two or more overlapping solids or regions.
- **SLICE (SLICE):** Divides a 3D solid or surface into two based on a cutting plane.

Annotation Commands:

- **DIMLINEAR (DLI):** Creates a linear dimension.
- **DIMALIGNED (DAL):** Creates an aligned dimension.
- **DIMANGULAR (DAN):** Creates an angular dimension.
- **DIMRADIUS (DRA):** Creates a radius dimension.
- **DIMDIAMETER (DDI):** Creates a diameter dimension.
- **DIMARC (DAR):** Creates an arc length dimension.
- **DIMBASELINE (DBA):** Creates a baseline dimension.
- **DIMCONTINUE (DCO):** Creates a continued dimension.
- **LEADER (LE):** Creates a leader line with annotation.
- **TOLERANCE (TOL):** Creates geometric tolerances.
- **STYLE (ST):** Creates and modifies text styles.
- **DDEDIT (ED):** Edits single-line text, dimension text, or attribute definitions.
- **MLEDIT (MLEDIT):** Edits multiline text.
- **TABLE (TABLE):** Creates a table object.

Block and External Reference Commands:

- **BLOCK (B):** Creates a block definition from selected objects.
- **INSERT (I):** Inserts a block or drawing into the current drawing.

- **WBLOCK (W):** Writes objects to a new drawing file (creates an external block).
- **XREF (XR):** Opens the External References palette to manage linked drawings.
- **XATTACH (XA):** Attaches an external reference (xref) to the current drawing.
- **XCLIP (XC):** Defines a clipping boundary for an external reference or block.
- **BEDIT (BE):** Opens the Block Editor to modify a block definition.
- **ATTDEF (ATT):** Defines attributes for a block.
- **ATTEDIT (ATE):** Edits attribute values in a block instance.
- **BATTMAN (BATTMAN):** Opens the Block Attribute Manager.

Layer Commands:

- **LAYER (LA):** Opens the Layer Properties Manager.
- **LAYERSTATE (LAS):** Saves, restores, and manages layer settings.
- **LAYISO (LAYISO):** Isolates selected layers.
- **LAYUNISO (LAYUNISO):** Restores all previously isolated layers.
- **LAYOFF (LAYOFF):** Turns off selected layers.
- **LAYON (LAYON):** Turns on all layers.
- **LAYFREEZE (LAYFRZ):** Freezes selected layers.
- **LAYTHAW (LAYTHW):** Thaws selected layers.
- **LAYLOCK (LAYLCK):** Locks selected layers.
- **LAYUNLOCK (LAYULK):** Unlocks selected layers.
- **MATCHPROP (MA):** Copies properties from one object to another.

View and Display Commands:

- **ZOOM (Z):** Increases or decreases the magnification of the view.
- **PAN (P):** Moves the view without changing the magnification.
- **VIEW (V):** Saves and restores named views.
- **VPORTS (VP):** Creates and manages viewport configurations in layouts.
- **REGEN (RE):** Regenerates the drawing and updates the display.
- **REGENALL (REA):** Regenerates all viewports in the drawing.
- **3DORBIT (3DO):** Enables interactive 3D viewing.
- **SHADE (SH):** Applies shading to 3D objects.
- **RENDER (R):** Creates a photorealistic image of a 3D model.

Utility Commands:

- **UNITS (UN):** Sets the drawing units.
- **OPTIONS (OP):** Opens the AutoCAD Options dialog box for customizing settings.
- **PROPERTIES (PR or CH):** Opens the Properties palette to view and modify object properties.
- **MEASURE (MEA):** Measures distance, area, and other properties.
- **ID (ID):** Displays the coordinates of a specified point.
- **DIST (DI):** Measures the distance and angle between two points.
- **AREA (AA):** Calculates the area and perimeter of objects or defined areas.
- **LIST (LI):** Displays information about selected objects.
- **CAL (CAL):** Opens the AutoCAD calculator.
- **DWGPROPS (DWGPROPS):** Opens the Drawing Properties dialog box.
- **AUDIT (AUDIT):** Evaluates the integrity of a drawing file.
- **RECOVER (RECOVER):** Attempts to recover a damaged drawing file.
- **PURGE (PU):** Removes unused named objects (layers, blocks, etc.) from the drawing.

System and Interface Commands:

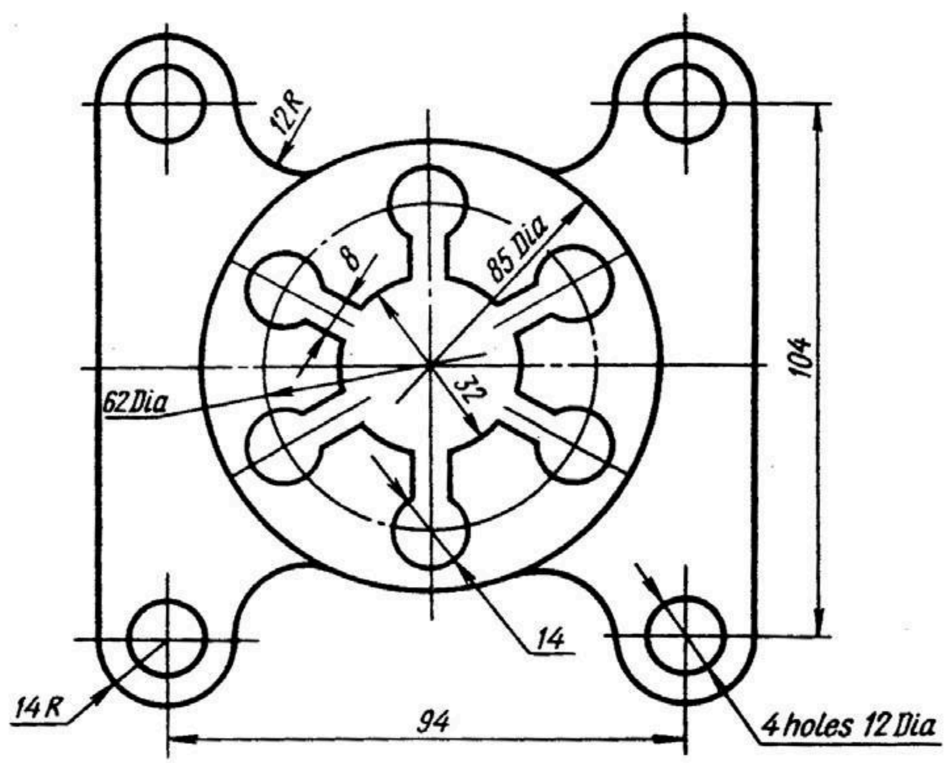
- **COMMANDLINE (CLI):** Displays or hides the command line window.
- **TOOLBAR (TOOLBAR):** Displays or hides toolbars.
- **MENULOAD (MENULOAD):** Loads a custom menu file.
- **CUI (CUI):** Opens the Customize User Interface editor.
- **QSAVE (QSAVE):** Saves the current drawing quickly.
- **SAVEAS (SAVEAS):** Saves the current drawing with a new name or format.
- **OPEN (OPEN):** Opens an existing drawing file.
- **NEW (NEW):** Creates a new drawing.
- **PLOT (PLOT or PRINT):** Opens the Plot dialog box for printing.
- **EXIT (QUIT):** Closes AutoCAD.

LAB EXPERIMENT-5

Aim: To create a 2D view of the given diagram using Auto CAD.

Procedure:

1. Type limits in command menu & set values to 45,45
2. Change the units to millimeters from inches and also precision to 0 by clicking format -> units -> ok.
3. To set the paper size type zoom -> enter and type a -> enter in command bar
4. Draw the 3 concentric circles of diameters 85 ,62,32
5. Draw the 2 axes lines from the centre of the circles
6. Draw the circle with 14dia on 62dia of circle and offset of the vertical line with distance 4 to both sides of the vertical line
7. Then trim the unwanted lines
8. Use the array command from modify tool bar to represent the 6 holes with 14 dia of centre of the circles
9. Offset the vertical and horizontal axes with 47 and 52 distance
10. And draw the 2 circles with 14 radius and 12 dia at coincide of the offset axes
11. From the modify tool bar select the fillet command to represent the 12R fillet
12. Then mirror this to require the 2D drawing
13. Finally trim the unwanted lines and circles



Cover

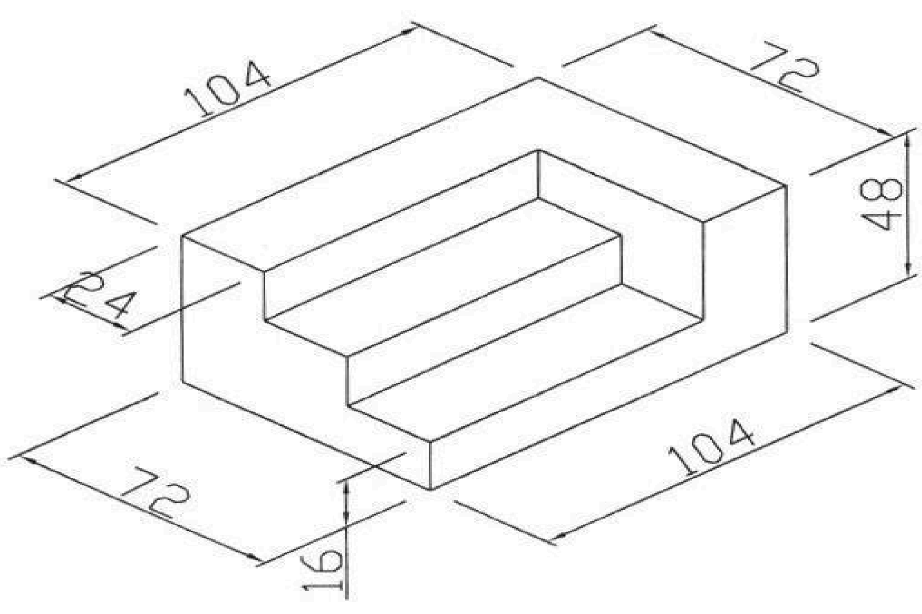
CAD Result: Hence the required 2D diagram is created using Auto.

LAB EXPERIMENT-6

Aim: To create a 2D isometric view of the given diagram using Auto CAD

Procedure:

1. Type limits in command menu & set value to 297,290.
2. Change the units to millimeters from inches and also precision to 0 by clicking format -> units -> ok.
3. To set the paper size type zoom -> enter and type a -> enter in command bar.
4. Go to drafting settings and turn on isometric snap..
5. Use the F5 key to change between the views of isometric planes.
6. Start from the front view and draw the the line of length of line 104 using the F8 key (O snap key) and continue with the 48 length line.
7. Change to top plane and draw the 72mm line.
8. Continue in the same fashion to complete the whole figure.
9. Give the dimensions from the dimension tool bar as in diagram.



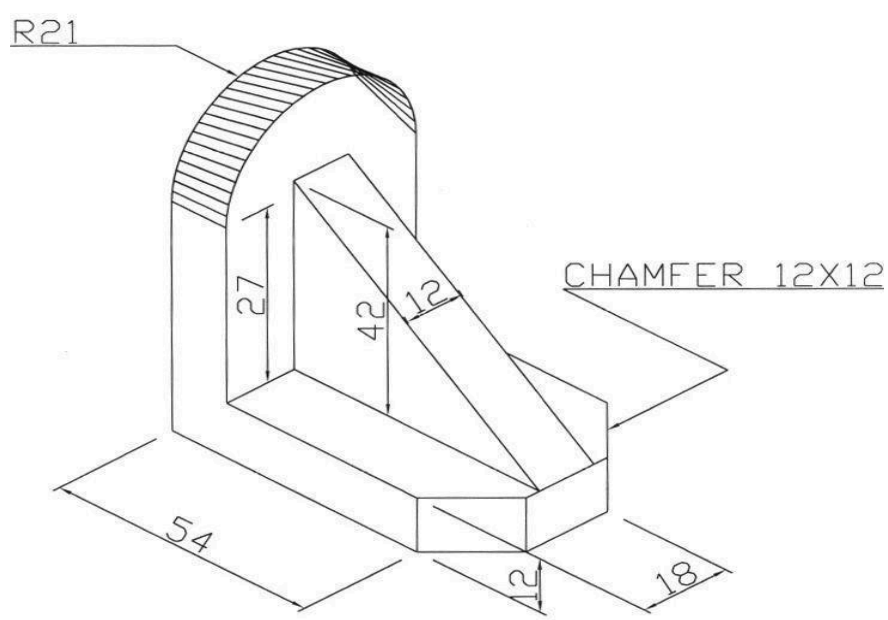
Result: Hence the required 2D isometric diagram is created using Auto CAD.

LAB EXPERIMENT-7

Aim: To create a 2D view of the given diagram using Auto CAD

Procedure:

1. Type limits in command menu & set value to 297,290.
2. Change the units to millimeters from inches and also precision to 0 by clicking format -> units ->ok.
3. To set the paper size type zoom -> enter and type a -> enter in Command bar.
4. Go to drafting settings and turn on isometric snap. Use the F5 key to change between the views of isometric planes.
5. Start from the front view and draw the the line of length of line 54.
6. Draw the semi circle using the Iso circle option from the ellipse command.
7. Continue drawing using F5 and F8 snap keys.
8. Give proper dimensions to the figure and practice at home



Result: Hence the required 2D isometric diagram is created using Auto CAD.